Yiding Wang

Electrical Engineering
University of Sydney

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LinkedIn Profile

Education

•Master of Engineering (Power) *2023-2024*
University of Sydney GPA: 75/100 (Distinction)

•Bachelor of Electrical Engineering (IoT) *2018-2022*
University of Sydney Second Class Honours

Experience

•Hubei Business College *2025.11 - Present*
Teaching Assistant Wuhan

- Instructed practical training courses including Power Electronics Technology, Electrical Control Systems, UAV Aerial Photography, Python Programming, and Industrial Robot Installation Commissioning.
- Delivered lectures to diverse student cohorts, including undergraduate, college-to-university transfer, and vocational college students.
- Achieved exceptional teaching outcomes, with student ratings for several courses consistently exceeding the course, faculty, and even university averages, as reflected in teaching quality feedback reports.

•Tutor Lim *2024.6 - Present*
Undergraduate Engineering Tutor Sydney

- Designed and delivered customized tutoring modules for 10+ undergraduate students, covering SoC design and IoT architecture.
- Led 5 end-to-end IoT projects, guided FPGA programming with Vivado; HLS workflows reduced development time by 30
- Improved average student grades by 15

•China Star Optoelectronic Technology (CSOT) *2020.10 - 2021.4*
Assistant Engineer, Lean Improvement Department Wuhan

- Orchestrated cross-departmental engineering training for 50+ employees, enhancing technical capabilities.
- Built university "Order Class" programs with 3 colleges, increasing new hire qualification and retention by 20
- Developed automated onboarding modules, accelerating new hire time-to-productivity by 20
- Coordinated logistics for expatriate technicians with strong cross-cultural communication.

Personal Projects

•Renewable Energy Microgrid Investment Optimization Model *2024*
Optimization Model & Algorithm Development University of Sydney

- Developed predictive optimization model to maximize ROI for renewable energy microgrids.
- Engineered hybrid BFGS + MIQP algorithm, improving efficiency by 20
- Enhanced prediction accuracy by 20
- Delivered annual cost savings of $500K for real-world microgrid projects.

•Electric Vehicle Charging Station Location Optimization *2022*
Data-Driven Siting Model University of Sydney

- Built optimization model for EV charging infrastructure economics and accessibility.
- Implemented PSO algorithm in Python, outperforming GA and SA in convergence.
- Integrated OpenStreetMap data, improving accessibility by 30

Technical Skills

Languages: Python, C, R
Tools: MATLAB, Simulink, Vivado, PSSE, MS Office
Professional Certifications: Theoretical Certificate in Safe Operation of Light Civil Unmanned Aerial Vehicles (CAAC)
Specialized Skills: Industrial Robot Installation & Commissioning, Electrical Control Systems Design and Troubleshooting
Language Proficiency: English (IELTS 7.5, Fluent), Mandarin (Native)
Soft Skills: Cross-cultural Communication, Project Management, Problem-Solving
Coursework: Power System Analysis, Power Electronics, Control Systems, Machine Learning
Interests: Renewable Energy, Microgrid, EV Infrastructure, Optimization

Achievements

- **First Prize, Smart City Design Competition** — *2023*
  Electrical System Lead — University of Sydney
  - Led 5-member cross-disciplinary team to design sustainable smart city energy solution.
  - Architected hybrid solar-wind-tidal energy system with storage for stable urban power supply.
- **High-Score IoT Infrastructure Monitoring System** — *2022*
  Full-Stack Development — University of Sydney
  - Built agricultural monitoring system with Fire Beetle, sensors, and MQTT protocol.
  - Realized anomaly alerts and auto-irrigation with 95